

L2: Co-citation and Bibliographic Coupling

The previous one-mode projections can also be computed using the adjacency matrix A that represents the bipartite graph.

Suppose that the element (i,k) of A is 1 if there is an edge from i to k – and 0 otherwise.

The co-citation metric $C_{i,j}$ for two nodes i and j is the number of nodes that have outgoing edges to both i and j . If i and j are items, then the co-citation metric is the number of users that have purchased both i and j .

On the other hand, the bibliographic coupling metric $B_{i,j}$ for two nodes i and j is the number of nodes that receive incoming edges from both i and j . If i and j are users, then the bibliographic coupling metric is the number of items that have been purchased by both i and j .

Co-citation of i and j	Bibliographic Coupling of i and j
$C_{ij} = \sum_{k=1}^n A_{ki} A_{kj} = \sum_{k=1}^n A_{ik}^T A_{kj}$	$B_{ij} = \sum_{k=1}^n A_{ik} A_{jk} = \sum_{k=1}^n A_{ik} A_{kj}^T$
$C = A^T A$	$B = A A^T$

As you can see both metrics can be computed as the product of A and A^T – the only difference is the order of the matrices in the product.